

Master's Thesis

Multi-objective Optimisation of suction-cup and two finger grippers using **todo: <Algorithms>**

im Studiengang Automotive Systems
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todo: <Semester>

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Betreuer: Daniel Schiller

Eidesstattliche Erklärung

Hiermit versichere ich, die vorliegende Arbeit selbstständig und unter ausschließlicher Verwendung der angegebenen Literatur und Hilfsmittel erstellt zu haben.

Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Esslingen, den 19. März 2026

Unterschrift

Sperrvermerk

Die nachfolgende Master's Thesis enthält vertrauliche Daten der Fraunhofer IPA. Veröffentlichungen oder Vervielfältigungen dieser Arbeit – auch nur auszugsweise – sind ohne ausdrückliche Genehmigung der Fraunhofer IPA nicht gestattet. Diese Arbeit ist nur den Prüfern sowie den Mitgliedern des Prüfungsausschusses zugänglich zu machen.

Zitat

„Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.“

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

„Aushängeschild“ der Arbeit, max 1 Seite

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1 Milestone Timeline

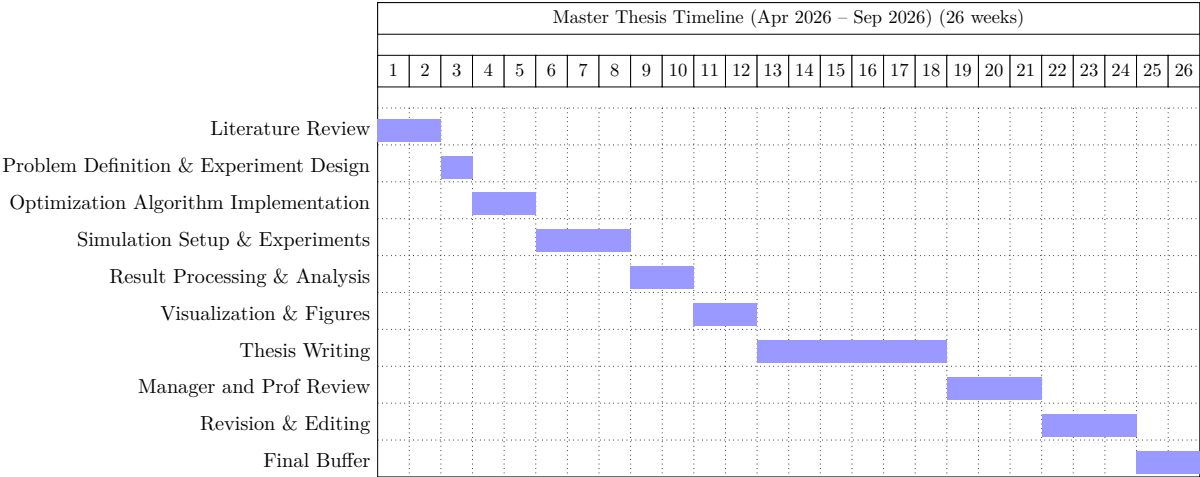


Abb. 1.1: Master thesis milestone plan

2 Einleitung

Die Gliederung des Hauptteils dieses Dokuments ist nur als Platzhalter gedacht. Im Folgenden sind zwei Vorschläge für Gliederungen dargestellt, die aber stets am konkreten Fall überprüft und in der Regel angepasst werden müssen.

Bitte beachten: Umfangreiche Programmlistings gehören nicht in die Arbeit, sondern auf eine beigefügte CDROM. Programmbeispiele können auszugsweise in der Arbeit gelistet werden, wenn sie zum Verständnis notwendig sind.

2.1 Literaturarbeit

1. Überblick (oder: Zusammenfassung, „Executive Summary“, alles Wichtige für den „Manager“ oder Schnelleser)
2. Fragestellung (oder: Ziele, Ausgangspunkt, Motivation)
3. Übersicht über den Stand der Wissenschaft und Technik (Beschreibung der Lösungsansätze, Beispiele etc. in einzelnen Abschnitten)
4. Bewertung der einzelnen untersuchten Ansätze, Beispiele etc., Identifikation von Defiziten
5. Synthese: Erstellung einer Gesamtschau, allgemeine Prinzipien, Beschreibung einer eigenen Sicht auf das Problem, evtl. auch eigene Vorschläge
6. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: eventuell recherchierte Texte, Produktbeschreibungen, etc.

2.2 System- und Softwareentwicklungen

1. Überblick (oder: Zusammenfassung, „Executive Summary“, alles Wichtige für den „Manager“ oder Schnelleser)
2. Problemstellung (oder: Ziele, Ausgangspunkt), Vorgesehener Benutzerkreis, Bedürfnisse der Benutzer
3. Stand der Technik (Wie wird das Problem bisher gelöst, wo sind die Defizite)
4. Gewählter Lösungsansatz (allgemeines Prinzip, welche Werkzeuge, z.B. Programmiersprachen werden verwendet)
5. Beschreibung der durchgeführten Arbeiten

6. Ergebnis (z.B. Screenshots mit Erläuterungen, Bedienanleitung)

7. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: evtl. (ausgewählte) Programmbeispiele.

3 Introduction

3.1 Motivation

Robotic manipulation plays a key role in modern automated production systems, particularly in logistics, assembly, and bin-picking applications. In such systems, the gripper forms the physical interface between the robot and the manipulated objects, directly influencing the reliability and efficiency of the manipulation process. Among the most commonly used gripper technologies are suction-cup grippers and two-finger mechanical grippers, each offering specific advantages depending on the properties of the objects to be handled.

Two-finger grippers provide versatile grasping capabilities for objects with varying geometries and are widely used in industrial manipulation tasks. Suction-cup grippers, on the other hand, are particularly effective for objects with smooth surfaces and enable fast and simple grasping operations with minimal mechanical complexity. However, determining suitable grasp configurations and gripper parameters for different tasks remains a challenging problem.

The performance of a gripper depends on multiple design and operational parameters such as finger geometry, gripping forces, actuator limits, and tool center point (TCP) positioning. These parameters are typically defined in configuration files describing the physical and functional properties of the gripper. For example, in the case of a two-finger gripper, parameters such as finger dimensions, actuator stroke, maximum gripping force, and TCP pose influence the resulting grasp quality. Similarly, suction-cup grippers depend on parameters including suction cup diameter, force limits, and allowable rotational deviation.

Traditional parameter tuning approaches often rely on heuristic rules or manual adjustments. Such approaches are time-consuming and do not guarantee optimal solutions, especially when multiple performance criteria must be considered simultaneously. In practical robotic applications, different objectives must be satisfied concurrently, including grasp stability, force limits, geometric feasibility, and manipulation efficiency.

Multi-objective optimization techniques provide a systematic approach to solving such problems by exploring the parameter space and identifying optimal trade-offs between competing objectives. Modern optimization algorithms such as Particle Swarm Optimization (PSO), Bayesian optimization methods, and Evolutionary Algorithms have demonstrated strong performance in solving complex, nonlinear optimization problems common in robotics.

Therefore, applying multi-objective optimization to the parameterization of suction-cup and two-finger grippers can significantly improve grasp planning and the overall performance of robotic manipulation systems.

3.2 Goal and Task of the Thesis

The goal of this thesis is to develop and evaluate a multi-objective optimization framework for determining optimal configurations of suction-cup and two-finger robotic grippers.

The grippers are defined through configuration files containing multiple adjustable parameters describing their mechanical properties, geometric dimensions, and operational limits. These parameters form the **decision variables** of the optimization problem. For the two-finger gripper, the configurable parameters include properties such as the number of fingers, actuator stroke limits, gripping effort, finger dimensions, and the position and orientation of the tool center point (TCP). In the case of the suction-cup gripper, relevant parameters include suction cup diameter, cup height, allowable rotation limits, and force limits such as coaxial, shear, and peeling forces.

The objective of the optimization process is to determine parameter combinations that result in improved grasp performance according to multiple criteria.

To achieve this goal, the following tasks are defined:

1. **Literature Review**

Study existing research on robotic grasping, gripper design, and multi-objective optimization techniques applied in robotic manipulation and grasp planning.

2. **Modeling of Gripper Configurations**

Analyze and model the configurable parameters of suction-cup and two-finger grippers based on their configuration files and identify the relevant decision variables for the optimization process.

3. **Definition of Objective Functions**

Define multiple objective functions representing different aspects of grasp quality, such as grasp stability, force feasibility, collision avoidance, and geometric constraints.

4. **Implementation of Optimization Algorithms**

Implement and apply several optimization algorithms, including Particle Swarm Optimization (PSO), Bayesian-based optimization methods, and Evolutionary Algorithms.

5. **Evaluation and Comparison**

Evaluate the performance of the different optimization methods with respect to convergence behavior, computational efficiency, and the quality of the obtained solutions.

6. **Analysis of Optimization Results**

Analyze the resulting Pareto-optimal solutions and identify suitable parameter configurations for both gripper types.

3.3 Approach

The grasp configuration problem is formulated as a multi-objective optimization problem in which several criteria relevant to grasp performance are optimized simultaneously. The adjustable parameters defined in the gripper configuration files serve as the decision variables of the optimization problem.

For the two-finger gripper, the decision variables include parameters related to the mechanical design and actuation system, such as actuator stroke limits, maximum gripping effort, finger geometry parameters (e.g., fingertip width and length), and the pose of the tool center point. For the suction-cup gripper, the decision variables include parameters such as suction cup diameter, cup height, rotation limits, and force limits including coaxial, shear, and peeling forces.

Based on these parameters, an optimization framework is established to evaluate different gripper configurations. Multiple objective functions are defined to quantify the quality of a given configuration according to criteria such as grasp feasibility, force limits, and geometric constraints.

To explore the search space efficiently, three different classes of optimization algorithms are applied:

- **Particle Swarm Optimization (PSO)**, a population-based optimization method inspired by collective behavior observed in natural systems.
- **Bayesian-based optimization methods**, which use probabilistic surrogate models to guide the search towards promising regions of the solution space.
- **Evolutionary Algorithms (EAs)**, which apply principles of natural evolution such as selection, mutation, and recombination to iteratively improve candidate solutions.

Each algorithm generates and evaluates candidate parameter configurations of the grippers. Since multiple objectives are considered simultaneously, the optimization process results in a set of **Pareto-optimal solutions** representing different trade-offs between the defined objectives.

The resulting solutions are analyzed and compared in order to evaluate the effectiveness of the different optimization approaches and to identify parameter configurations that improve the grasp performance of suction-cup and two-finger grippers.

4 Methodology

4.1 Methodology for Multi-Suction Cup Gripper Optimization

This section describes the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration.

4.1.1 Parameterization of the Gripper

Each suction cup is defined by its planar position, while the height and orientation remain fixed. Thus, the decision variables are:

$$\mathbf{p} = [(x_1, y_1), (x_2, y_2), (x_3, y_3)] \quad (4.1)$$

Each individual $\mathbf{p}$ forms a triangular base of the gripper.

Assumption In this work, a triangular configuration of suction cups is assumed to yield a stable grasp. This is motivated by the fact that three spatially distributed contact points form a statically stable support, enabling effective load distribution and resistance to external disturbances.

4.1.2 Initialization

An initial population of candidate solutions is randomly generated:

$$\mathcal{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N\} \quad (4.2)$$

4.1.3 Workpiece Representation

Each workpiece is represented as a point cloud (PCD), where each point corresponds to a potential grasp location.

4.1.4 Optimization Loop

The optimization proceeds iteratively. For each individual $\mathbf{p} \in \mathcal{P}$, a subprocess is executed to evaluate grasp quality.

Subprocess: Grasp Evaluation

For each individual $\mathbf{p}$, the following steps are performed:

1. Grasp Generation

For each point in the workpiece point cloud, a candidate suction grasp is generated by aligning the tool center point (TCP) of the gripper with the surface point.

2. Collision Filtering

The gripper is modeled as a rigid 3D body with fixed height, forming a triangular prism whose base is defined by $\mathbf{p}$.

Each candidate grasp is checked for collisions between the gripper body and the workpiece:

- If a collision occurs, the grasp is discarded.
- Otherwise, the grasp is considered valid.

This step filters out physically infeasible grasps.

3. Clustering of Valid Grasps

Let $\mathcal{G}_{valid}$ denote the set of valid grasps.

A fixed number of clusters K is defined as a hyperparameter.

- If $|\mathcal{G}_{valid}| < K$, clustering is skipped.
- Otherwise, clustering is performed based on spatial proximity and orientation similarity.

This step reduces redundancy and groups similar grasps.

4. Triangle Area Calculation

The area of the triangle formed by the suction cups is computed as:

$$A(\mathbf{p}) = \frac{1}{2} |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)| \quad (4.3)$$

5. Diversity Metric Calculation

A diversity score $D(\mathbf{p})$ is computed based on the spatial distribution of the clustered grasps:

- Distances between cluster centers are calculated and normalized.
- Optionally, orientation differences between grasps are included.

The diversity metric encourages wide coverage of the workpiece surface.

Objective Evaluation

After the subprocess, each individual $\mathbf{p}$ is assigned objective values:

$$\mathbf{f}(\mathbf{p}) = [-A(\mathbf{p}), -D(\mathbf{p})] \quad (4.4)$$

where:

- $A(\mathbf{p})$ is maximized (larger triangle improves stability)
- $D(\mathbf{p})$ is maximized (greater diversity improves robustness)

Population Update

The evaluated individuals are passed to a multi-objective optimization algorithm, which generates a new population:

$$\mathcal{P}^{(t+1)} = \text{Optimizer}(\mathcal{P}^{(t)}, \mathbf{f}) \quad (4.5)$$

The process is repeated until a termination criterion is satisfied.

4.1.5 Hyperparameters

The framework includes several predefined hyperparameters:

- Population size N
- Number of clusters K
- Maximum number of iterations

These parameters influence the convergence behavior and computational cost of the optimization.

4.1.6 Output

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{p} \mid \mathbf{p} \text{ is Pareto-optimal}\} \quad (4.6)$$

These solutions represent optimal trade-offs between triangle area and grasp diversity.

5 Stand der Technik

Beschreibung des momentanen Standes der Technik und Wissenschaft. Hier muss auch klar definiert werden, in welchem Zusammenhang das Problem steht, d.h. der Systemkontext und der bekannte Lösungsraum ist klar zu beschreiben (vorhandene.Lösungen

5.1 Problembeschreibung

5.2 Existierende Lös¹ungen

5.3 Literaturverweise

Verweise im Text: [1] und [Gun04].

6 Lösungsansatz

Hier können je nach Art der Arbeit auch verschiedene Lösungsansätze beschrieben und diskutiert werden.

7 Durchgeführte Arbeiten

Es muss klar erkennbar sein, auf was aufgesetzt wurde, d.h. schon vorhanden war, und was im Rahmen dieser Arbeit selbst entwickelt wurde.

8 Ergebnisse

Hier kann z.B. eine Bedienanleitung stehen, oder bei Analysen die Analyseergebnisse. „Neuigkeiten“ Messergebnisse

9 Zusammenfassung

Kurze Zusammenfassung und evtl. Beschreibung der Punkte, die zu einer kompletten, marktreifen Lösung noch fehlen.

10 Schluss

Ergebnis-Bewertung, Zusammenfassung und Ausblick

A Kapitel im Anhang

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

Literaturverzeichnis

- [1] Thomas Nonnenmacher, LaTeX Grundlagen - Setzen einer wissenschaftlichen Arbeit Skript, 2008,<http://www.stz-softwaretechnik.de>; (*Bei STZ Internetseite unter Publikationen - Skripte*) [V. 2.0 26.02.08]
- [Gun04] Karsten Günther, LaTeX2 — Das umfassende Handbuch, Galileo Computing, 2004, <http://www.galileocomputing.de/katalog/buecher/titel/gp/titelID-768>; 1. Auflage